Synoptic CB: Macrophyte Fluxes

2022 Samples

2026-05-11

Contents

#Read in and collate all the files in the Processed Data folder

```
ch4 <- read.csv("Processed Data/COMPASS_Synoptic_CB_Macrophyte_Fluxes_Final_2022_CH4.csv")
ch4$X <- NULL
ch4 <- ch4 %>%
  rename_with(
    .cols = 7:(ncol(.) - 4),
    .fn = ~paste0("ch4_", .)
  ) %>%
  rename(ch4_qaqc_flag = qaqc_flag)

co2 <- read.csv("Processed Data/COMPASS_Synoptic_CB_Macrophyte_Fluxes_Final_2022_CO2.csv")
co2$X <- NULL
co2 <- co2 %>%
  rename_with(
    .cols = 7:(ncol(.) - 4),
    .fn = ~paste0("co2_", .)
  ) %>%
  rename(co2_qaqc_flag = qaqc_flag)

# Combine all files
all_dat <- merge(ch4, co2, by = c("Plot", "Site", "Zone", "Replicate", "Date", "TIMESTAMP", "TIMESTAMP_r
```

##Pull in Hydrago probe data

```
## All flux IDs are present in hydrago data
```

##pull TEROS data for the zone during the time of the fluxes and put it into another column so that we can use that too.

```r
# Soil sensors

##Read in L1 Sensor Data from S Drive and only save necessary data
# Path to zip file
L2 <- "S:/Biogeochemistry/COMPASS/L2 Data/v2-1"

# List CSV files inside the zip
files <- list.files(path = L2, recursive = TRUE, full.names = TRUE)

files <- files[
  grepl("soil", files, ignore.case = TRUE) &
  grepl("10cm", files, ignore.case = TRUE) &
  grepl("SWH|GCW|MSM|GWI", files, ignore.case = TRUE) &
  grepl("2022", files, ignore.case = TRUE) &
  grepl("\\.parquet$", files, ignore.case = TRUE)
]

teros <- files %>%
  map_dfr(read_parquet)

#Change W Plot to WC
teros <- teros %>%
  mutate(Plot = ifelse(Plot == "W", "WC", Plot))

##Want to go from long to wide format
teros_wide <- pivot_wider(teros, id_cols = c("TIMESTAMP", "Plot", "Site", "Instrument", "Instrument_ID"

##Average for each TIMESTAMP by Site, Plot, and Timestamp
teros_wide_avg <- teros_wide %>%
  group_by(Site, Plot, TIMESTAMP) %>%
   dplyr::summarize(
     soil.EC.10cm.TEROS = mean(`soil-EC-10cm`, na.rm=TRUE),
     soil.salinity.10cm.TEROS = mean(`soil-salinity-10cm`, na.rm=TRUE),
     soil.temp.10cm.TEROS = mean(`soil-temp-10cm`, na.rm=TRUE),
     soil.vwc.10cm.TEROS = mean(`soil-vwc-10cm`, na.rm=TRUE))
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by Site, Plot, and TIMESTAMP.
## i Output is grouped by Site and Plot.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(Site, Plot, TIMESTAMP))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

```r
##Write out this dataframe
write.csv(teros_wide_avg, file="Raw Data/TEROS_data.csv", row.names = FALSE)

teros_wide_avg <- read.csv("Raw Data/TEROS_data.csv")
colnames(teros_wide_avg)
```

```
## [1] "Site"                     "Plot"
## [3] "TIMESTAMP"                "soil.EC.10cm.TEROS"
## [5] "soil.salinity.10cm.TEROS" "soil.temp.10cm.TEROS"
## [7] "soil.vwc.10cm.TEROS"
```

0.1 Match up the TEROS soil data and add it to the flux dataframe

```r
#Going to need to use site and timestamp in the metadata in order to find a date

#first we need to make a timestamp column in metadat so that we can
final_dat1$TIMESTAMP <- ymd_hms(final_dat1$TIMESTAMP)

#make sure time stamps are in the correct format
final_dat1$TIMESTAMP <- as.POSIXct(final_dat1$TIMESTAMP, format = "%Y-%m-%d %H:%M:%OS")
teros_wide_avg$TIMESTAMP <- as.POSIXct(teros_wide_avg$TIMESTAMP, format = "%Y-%m-%d %H:%M:%OS")

#time rounding step to closest 15 minutes
final_dat1$DateTime_round <- round_date(final_dat1$TIMESTAMP, "15 mins")
teros_wide_avg$TIMESTAMP_round <- round_date(teros_wide_avg$TIMESTAMP, "15 mins")

#ensure there is only one data observation per hour
teros_wide_avg_clean <- teros_wide_avg %>%
    group_by(TIMESTAMP_round, Site, Plot) %>%
    summarize(
     soil.EC.10cm.TEROS = mean(soil.EC.10cm.TEROS, na.rm=TRUE),
     soil.salinity.10cm.TEROS = mean(soil.salinity.10cm.TEROS, na.rm=TRUE),
     soil.temp.10cm.TEROS = mean(soil.temp.10cm.TEROS, na.rm=TRUE),
     soil.vwc.10cm.TEROS = mean(soil.vwc.10cm.TEROS, na.rm=TRUE))
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by TIMESTAMP_round, Site, and Plot.
## i Output is grouped by TIMESTAMP_round and Site.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(TIMESTAMP_round, Site, Plot))` for per-operation
##   grouping (`?dplyr::dplyr_by`) instead.
```

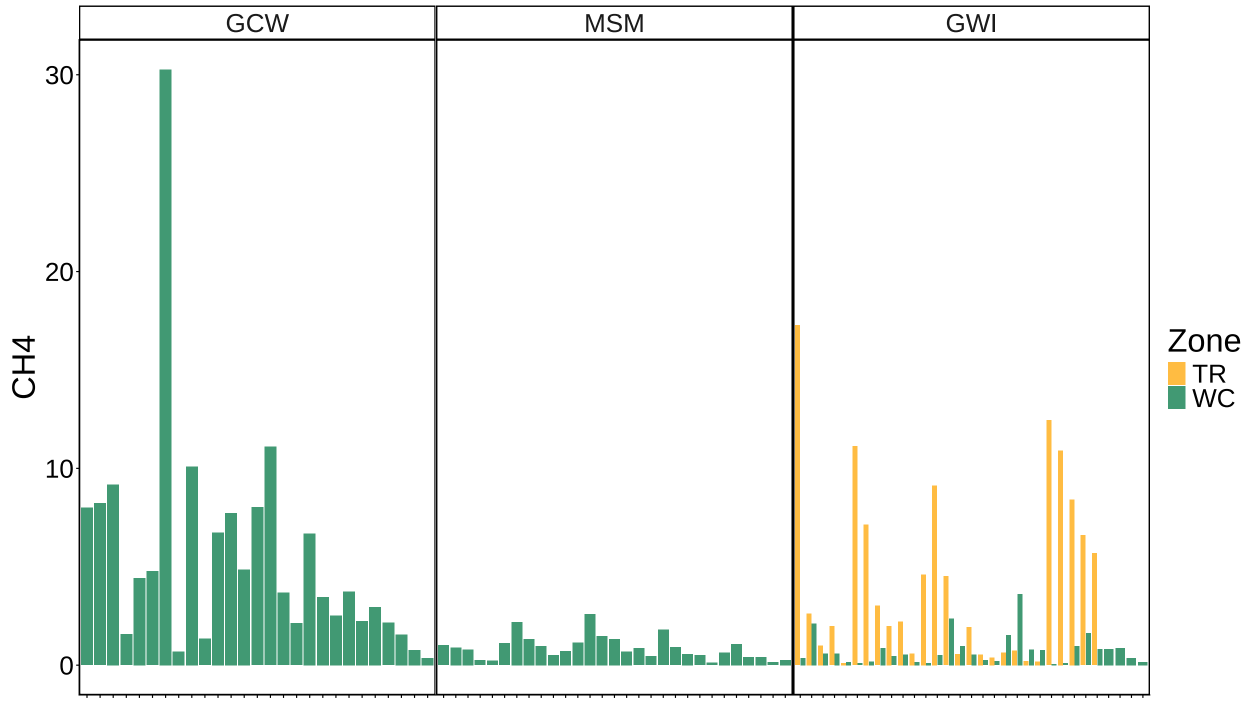
```r
#join them
final_dat2 <- final_dat1 %>%
  left_join(teros_wide_avg_clean, by=c("Site"="Site", "Zone"="Plot", "DateTime_round"="TIMESTAMP_round")
  select(-DateTime_round)
```

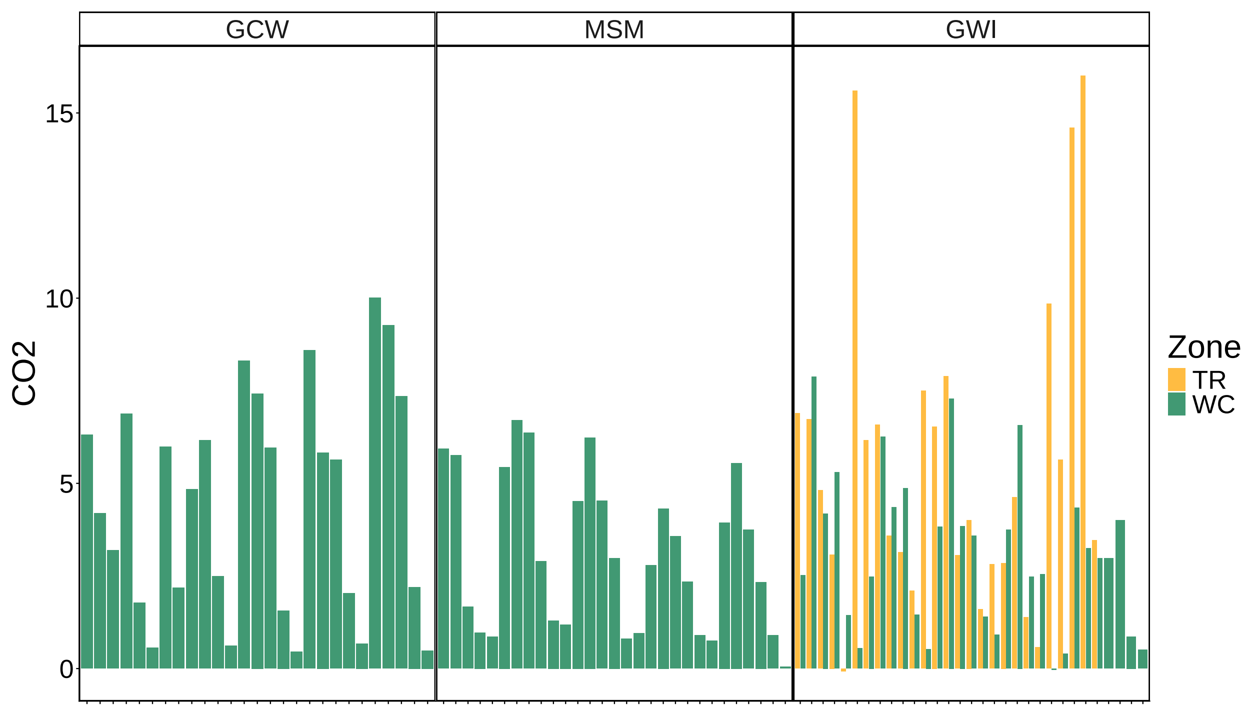
##Remove bad fluxes and calculate how many there were

```
## Number flux measurements removed: 2

## Number negative CO2 fluxes removed: 0

## Number messy fluxes removed: 2

## Percentage flux measurements removed: 1.74%

## Percentage negative CO2 fluxes removed: 0%

## Percentage messy fluxes removed: 1.74%
```

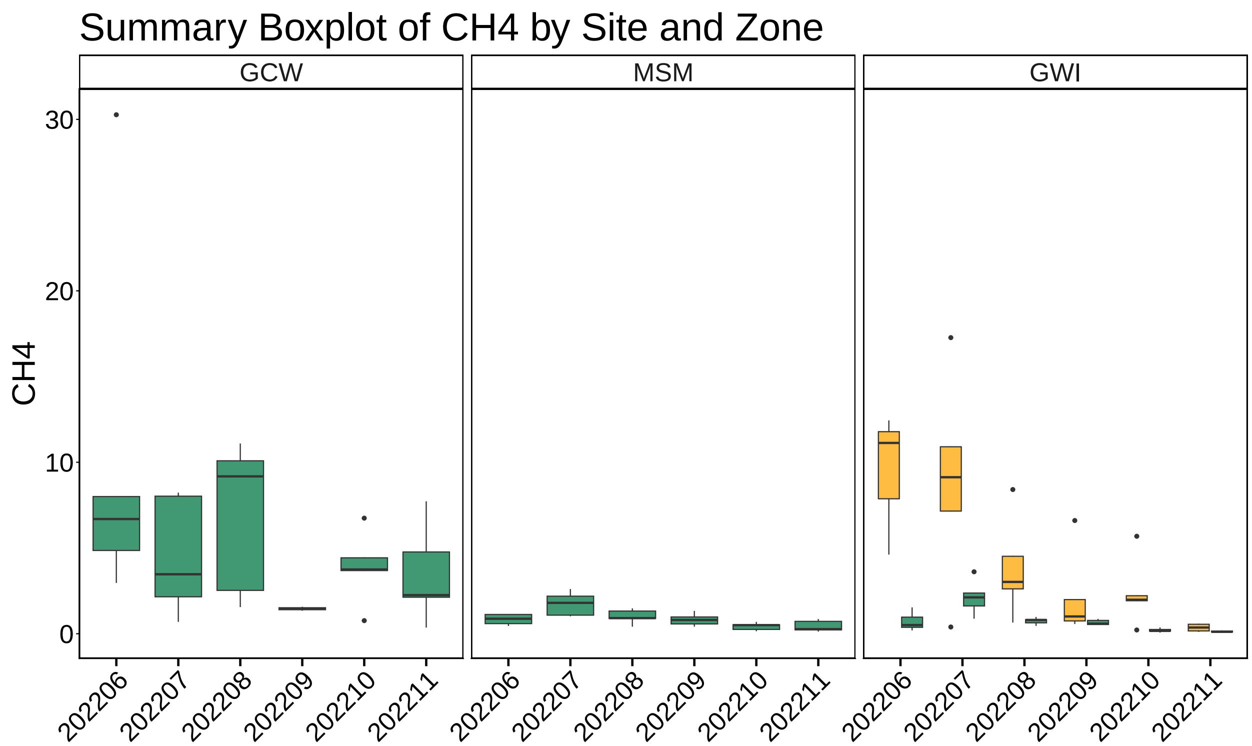
0.2 Write out files

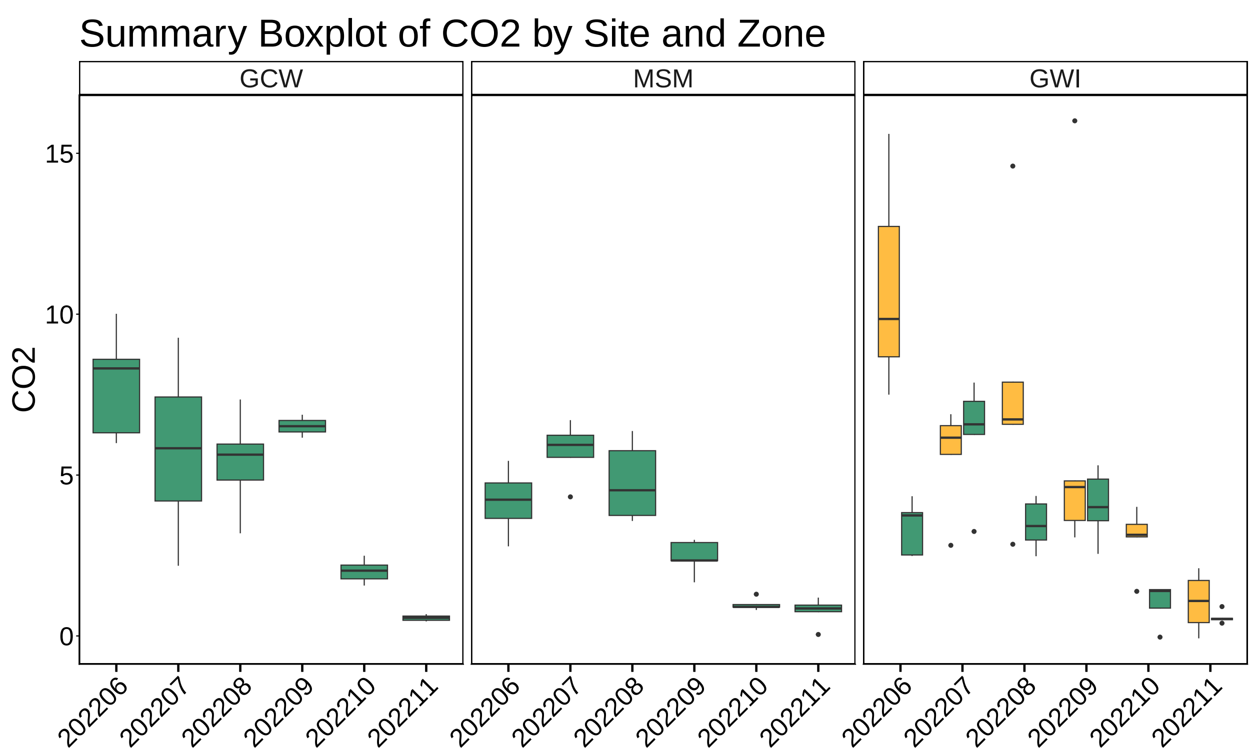
0.3 All Data by Year, Site and Zone



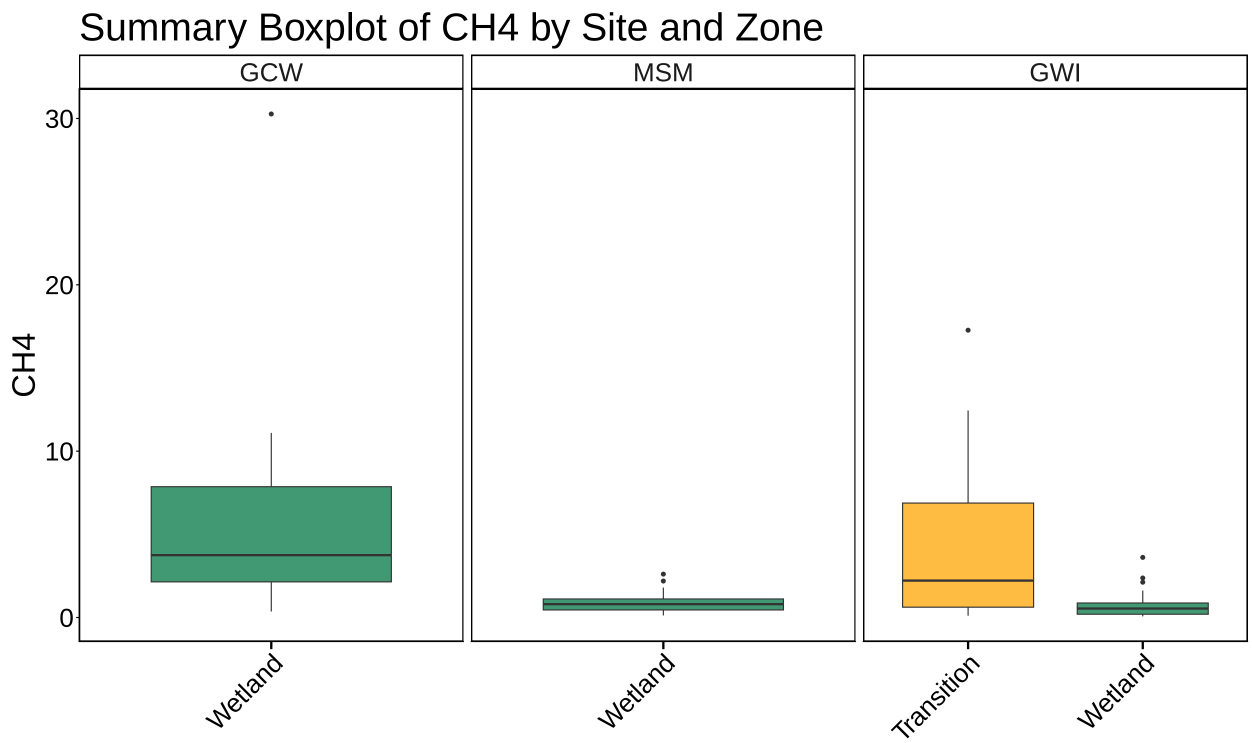


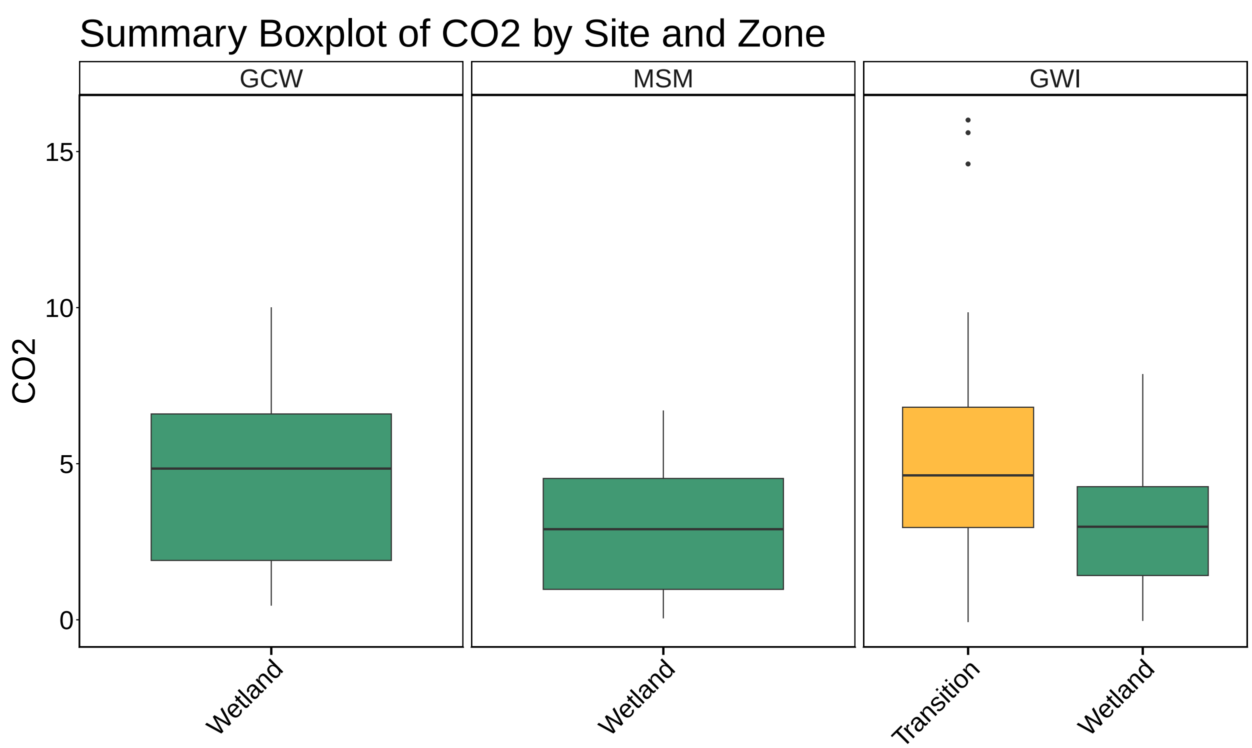
0.4 Boxplot summary by Date





0.5 Boxplot summary by Site





##Calculate summary metrics

0.6 Summary Tables

Table 1: Summary Statistics

Site	Zone_1	min_ch4	max_ch4	mean_ch4	min_co2	max_co2	mean_co2
GCW	Wetland	0.3585206	30.270376	5.5296620	0.4518196	10.011470	4.481874
MSM	Wetland	0.1222875	2.606256	0.8783515	0.0458841	6.706775	3.115307
GWI	Transition	0.1071858	17.271153	4.3189199	-0.0745288	16.009112	5.594259
GWI	Wetland	0.0653984	3.612942	0.7609969	-0.0364413	7.873652	3.139441